Yuyuan Han

Computational imaging & physics-aware learning

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I work on optical measurement design and learning for imaging and perception. Current research explores acquisition-aware learning and pretrained models for optical signals.

EDUCATION & RESEARCH APPOINTMENTS

Xi'an Jiaotong University · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

Northwest University · B.Eng. — 2018–2022
Electronic Information Science and Technology

Politecnico di Milano · Visiting PhD student — Oct 2026–present
Department of Physics · Host: Antonio Pifferi

SELECTED RESEARCH

The Lift Spectrum · Robust image-free perception

Preprint

arXiv · 2026 · Paper · Code

- Showed that segmenting directly from 3.13%-sampled single-pixel measurements beats the strongest clean-trained reconstruct-then-segment baseline by 3.1–21.7 foreground-mIoU points under 20 dB noise.
- Designed a content-adaptive measurement-to-space lift trained with scheduled auxiliary reconstruction, exceeding the prior image-free baseline by 3.2–9.9 points at the same sampling rate.

Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Designed tunable-sparsity Hadamard illumination with two acquisition modes; derived a sparsity-noise model and validated noise resilience in simulations and an optical experiment through ground glass.

LDGI · Optical phased-array imaging

Optics Communications · 2025 · Paper · Code

- Showed that line patterns from one-dimensional optical phased arrays break the noise symmetry assumed by traditional ghost-imaging denoising; developed a loop differential algorithm and validated it optically.

CURRENT RESEARCH

Developing acquisition-aware models and adapting pretrained multimodal models to bucket-detector signals. Planned visiting research: time-domain diffuse optics and 3D reconstruction.

ADDITIONAL EXPERIENCE

Photonic device modeling · Modeled and optimized electro-optic phased-array geometries for beam steering.

Optical system integration · Led mechanical design, optical alignment and system integration; participated in underwater field tests.

AWARD

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code

Yuyuan Han

PUBLICATIONS

Peer-reviewed journal articles

01 Chong Wang, Hao Lu, **Yuyuan Han**, Huaibin Zheng, Yuchen He. **Graph-Regularized Estimation for Inverse Problems in Computational Optical Imaging**. *IEEE Transactions on Image Processing* 35, 9314–9326 (2026). doi:10.1109/TIP.2026.3726365

02 Jingwei Li, **Yuyuan Han**, Aoxiang Chen, Pengpeng Li, Xinglin Zhu, Guanbao Zhao, Luohan Peng, Huaibin Zheng. **Optical Phased Arrays on Photonic Integrated Circuit Platforms: Challenges, Opportunities, and Future Applications**. *Nanophotonics* 15(14), e70182 (2026). doi:10.1002/nap2.70182

03 **Yuyuan Han**, Peng Wu, Hanxiong Xu, Bin Li, Yuchen He, Jianbin Liu, Hui Chen, Huaibin Zheng. **Generalized sparse Hadamard single-pixel Imaging**. *Optics Express* 33(20), 43324–43341 (2025). doi:10.1364/OE.576533

04 **Yuyuan Han**, Huaibin Zheng, Bin Li, Jingwei Li, Long Qiu, Wenxuan Hao, Zheng Dang, Hui Chen, Jianbin Liu, Yuchen He, Yanyan Liu, Zhuo Xu. **Loop differential ghost imaging based on line pattern**. *Optics Communications* 580, 131589 (2025). doi:10.1016/j.optcom.2025.131589

05 Jingwei Li, Huaibin Zheng, **Yuyuan Han**, Bin Li, Wenxuan Hao, Long Qiu, Yanyan Liu, Yuchen He, Xiaoyong Wei, Zhuo Xu. **Design and analysis of single-electrode integrated lithium niobate optical phased array for two-dimensional beam steering**. *Optics and Lasers in Engineering* 184, 108617 (2025). doi:10.1016/j.optlaseng.2024.108617

06 Long Qiu, Huaibin Zheng, **Yuyuan Han**, Bin Li, Wenxuan Hao, Zheng Dang, Yuanlin Yu, Jiawei Li, Hanxiong Xu, Hui Chen, Jianbin Liu, Yuchen He, Yanyan Liu, Zhuo Xu. **Fusion of near-infrared single-photon LiDAR with visible light camera**. *Measurement Science and Technology* 36(6), 065409 (2025). doi:10.1088/1361-6501/addddf

07 Huaibin Zheng, **Yuyuan Han**, Long Qiu, Yanfeng Zong, Jingwei Li, Yu Zhou, Yuchen He, Jianbin Liu, Gao Wang, Hui Chen, Zhuo Xu. **Long-Range Imaging LiDAR with Multiple Denoising Technologies**. *Applied Sciences* 14(8), 3414 (2024). doi:10.3390/app14083414

08 Yanfeng Zong, Huaibin Zheng, Xinwei Wu, Jingwei Li, Long Qiu, **Yuyuan Han**. **Sequence-Controlled Pseudothermal Optical Ghost Imaging System**. *Acta Optica Sinica* 43(7), 0711001 (2023). doi:10.3788/AOS221781

Preprints

09 **Yuyuan Han**, Jingwei Li, Xiaoxia Zhang, Long Qiu, Chong Wang, Wenxuan Hao, Jiangyu Han, Xinyu Yao, Yuchen He, Hui Chen, Jianbin Liu, Huaibin Zheng. **The Lift Spectrum: How Measurement-to-Space Adaptivity Shapes Robustness in Image-Free Single-Pixel Sensing**. *arXiv* (2026). doi:10.48550/arXiv.2607.22077

RESEARCH CODE

Comparison-of-SPI-DLs · Comparative evaluation of learning-based single-pixel reconstruction.

remote-gpu-trainer · Open-source agent skill for reproducible remote-GPU deep-learning experiments.